

An AR Visualization System for 3D Carbon Dioxide Concentration Measurement Using Fixed and Mobile Sensors

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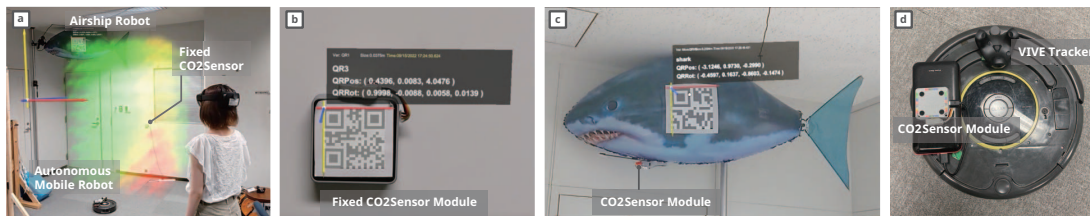


Figure 1: (a) Proposed AR visualization system for carbon dioxide concentration, (b) Fixed CO₂ sensor module, (c) Airship robot with a CO₂ sensor module, and (d) Autonomous mobile robot with a CO₂ sensor module.

ABSTRACT

Recently, the use of CO₂ concentration monitors as a guide for ventilation is increasing in spaces where many people gather, such as offices. However, 2D map-based visualizations are not enough to understand the progression of indoor air pollution. Therefore, we propose a three-dimensional (3D) visualization of CO₂ concentration using a head-mounted display (HMD). A 3D distribution of CO₂ concentration is automatically calculated using the position coordinates and measured values of fixed and mobile sensors. The results of the preliminary user evaluation suggested that AR visualization may be a more effective way to inform the need for ventilation than conventional methods.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Visualization—Visualization systems and tools—Visualization toolkits

1 INTRODUCTION

In indoor environments, keeping good ventilation practices is necessary to maintain healthy and comfortable air quality. Inadequate ventilation causes high carbon dioxide (CO₂) concentrations, which have adverse effects on the human body, including headache, drowsiness, carbon monoxide poisoning, and impaired concentration and memory [6]. Furthermore, viruses remain indoors for long periods of time, which increases the risk of viral infections [7].

The standard for appropriate ventilation recommends that CO₂ concentration should be kept below 1000 [ppm] [5]. Recently, the use of CO₂ concentration monitors is increasing in spaces where many people gather, such as offices. However, 2D map-based visualizations are not enough to understand the progression of indoor air pollution and whether there is a need for ventilation. Therefore, we

propose a 3D visualization of CO₂ concentration using an HMD to facilitate the understanding of CO₂ concentration in space.

Numerical fluid dynamics models such as CFD simulations are often used to analyze atmospheric gases. Chae et al. proposed an AR system where 3D mapping and air conditioner installation are made interactively, and then airflow/temperature simulation and visualization are made in real-time, unlike previous works that rely on pre-scanned indoor environments and pre-calculated CFD results [3]. However, a comparison of temperatures measured with 792 thermometers in the test room and their simulation results showed that the simulation was less accurate directly below the air conditioner and near the wall. In addition, Batoga et al. reported a discrepancy between the predicted CO₂ concentrations from CFD simulations and the actual sensor measurements [1]. Therefore, in this study, changes in CO₂ concentration in a real environment are measured and visualized in real-time. Previously, Burgués et al. proposed a method to visualize the temporal variation of 3D gas distributions generated by gas sources located at diverse locations [2]. To measure the 3D gas distribution, they divided the room into 3D grids, and sensors were placed in each grid. The results show that gas heavier than air does not accumulate near the ground surface and that the gas distribution changes drastically with time, regardless of height. However, conventional methods have some problems, such as requiring the installation of a large number of sensors and manual position measurement.

To address these issues, we aim to achieve high-density, real-time 3D measurement and visualization in real environments without requiring time-consuming manual labor. Therefore, we propose an augmented reality (AR) visualization system for indoor CO₂ concentration (see Fig. 1) using both fixed sensors and sensors mounted on mobile robots that have the following characteristics:

- Automatic localization of fixed CO₂ gas sensors in 3D.
- Automatic localization of CO₂ gas sensors mounted on mobile robots in 3D.
- Interpolation of the measurement to estimate CO₂ density distribution.
- AR visualization of CO₂ concentration distribution.

2 SYSTEM

In this study, we build an AR visualization system for CO₂ concentration using both fixed and mobile sensors. The system automatically

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calculates the distribution of CO₂ concentration in a room using the position coordinates and measured values of fixed sensors that are placed at arbitrary positions of interest and mobile sensors that patrol the room. The visualization model corresponding to the calculated CO₂ concentration is displayed in the form of AR in real-time using an AR-HMD. We believe that many people will be wearing AR glasses in the future. This system can be used for daily sensing of CO₂ concentration in offices and other indoor spaces. The system can also be used to stimulate ventilation behavior, help optimize furniture placement, and improve the layout of air conditioning equipment such as circulators.

2.1 Sensing

We implemented a prototype system using several CO₂ gas sensor modules fixed in various indoor locations and those mounted on an autonomous mobile robot (iRobot Roomba 600 Series) and an airship robot (William Mark Air Swimmer Shark). The CO₂ gas sensor modules consist of an M5Stack Gray (120 [g]) and a TVOC/eCO₂ gas sensor unit (SGP30). The CO₂ concentration is measured at 1 Hz. The minimum measurement value of the sensor is 400 [ppm]. The microcomputer for the CO₂ gas sensor module mounted on the airship robot is an M5Stick C Plus (21 [g]) to reduce weight.

2.2 Localization of sensors

AR markers (QR codes) attached to the CO₂ gas sensor modules are detected by the built-in camera of the HMD to obtain the position coordinates of fixed sensors placed at various locations in the room. We use a Microsoft HoloLens 2 as an AR-HMD. The positions of the mobile robots are tracked either by a QR code or by an HTC VIVE Tracker (for the autonomous robot). In the case of QR codes, the position can be continuously acquired while the user is looking at the QR code attached to the mobile robot. If a continuous measurement is desired while the user is not looking at the QR code, an external motion tracker can be attached to the autonomous mobile robot. CO₂ concentration distribution can then be calculated with higher accuracy. The position coordinates of the tracker in the coordinate system of the HMD are calculated using the transformation matrix and obtained automatically. The automatic sensor localization enables the automatic calibration of sensor placement.

2.3 Visualization of CO₂ concentration distribution

The visualization space is divided into voxel grids which are set around the initial position of the HoloLens 2, and the CO₂ concentration at each grid point is calculated by linear interpolation from the position coordinates and measurements of each sensor. By this decoupling, smooth visualization is realized regardless of the sensor density and layout. The weighted average method is used to estimate the value of each grid point from irregularly distributed measurements [4]. Measurement values by mobile sensors are decreased exponentially over time and are forgotten after a certain time.

We render a particle-based smoke-like visualization model based on the calculated CO₂ concentration. This method is considered semi-transparent volume rendering. It is similar to a traditional splatting technique, but instead of splatting, the smoke particles are assigned a color based on the CO₂ concentration and opacity. The size of each voxel can be decided on a case by case basis but is empirically set to 20 x 20 x 20 [cm]. Simply filling each voxel with a single color will result in noticeable jaggies. In addition, low transparency reduces the visibility of the real environment, while high transparency reduces the visibility of the visualization. For these reasons, we decided to use particles for visualization. The color of the smoke particles is continuously mapped to the CO₂ concentration, for example, green for low and red for high. The transfer function can be changed by a configuration file depending on the user's preference.

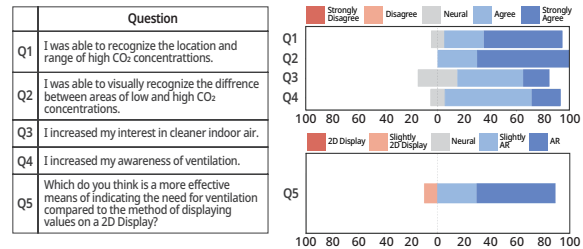


Figure 2: Likert scale assessment results.

3 USER EVALUATION

Ten participants in their twenties experienced the prototype system for about five to ten minutes. A 5-point Likert scale was adopted for the evaluation of this system. The distribution of the survey results is shown in Fig 2. From the responses collected for Q1, Q2, Q3, and Q4, it can be concluded that all four categories of the system showed positive trends, including visibility of high-concentration areas, visibility of differences in concentration, increased awareness of indoor air quality, and increased awareness of ventilation. The results of Q5 suggest that AR visualization may be a more effective way to inform the need for ventilation than conventional methods.

4 CONCLUSION AND FUTURE WORK

We have proposed an AR visualization system for indoor CO₂ concentration distribution using CO₂ sensor modules, mobile robots, and an AR-HMD. A preliminary evaluation revealed that the participants found the system useful and promising. Our system can be used as a platform to visualize not only CO₂ but also other gases by changing the sensors mounted on the measurement modules. We currently do not control the movement of mobile robots. It is necessary to implement an automatic patrol method for mobile robots that is compatible with an environment where humans coexist. In the future, we plan to evaluate the accuracy of linear interpolation under multiple conditions, such as the number of sensors used, and when measurements are taken with only fixed sensors or only mobile sensors. The color assignment of the visualization model is set to be heuristically easy to see, referring to the standard recommended by the national government of a country. However, since it is unclear whether this is the optimal color mapping, we plan to ask experts for their opinions. Another problem that is always present in volume rendering is that the depth of information is difficult to understand due to the information in the foreground. We are considering performing thresholding to display only the darker areas, or adding a fog effect to display only a certain amount of the nearby area.

REFERENCES

- [1] P. Batog *et al.* Dynamic of changes in carbon dioxide concentration in bedrooms. *Procedia Engineering*, 57:175–182, 2013.
- [2] J. Burgués *et al.* Gas distribution mapping and source localization using a 3d grid of metal oxide semiconductor sensors. *Sensors and Actuators B: Chemical*, 304:127309, 2020.
- [3] J. Chae *et al.* Virtual air conditioner's airflow simulation and visualization in ar. *ACM VRST '22*, 2022.
- [4] Y. Inui *et al.* Griview3d: Three dimensional gridding and visualization based on irregularly spaced observations. *Geoinformatics*, 20(4):197–210, 2009.
- [5] Ministry of Health, Labour and Welfare. About building environmental sanitation management standards, 2022.
- [6] U. Satish *et al.* Is co2 an indoor pollutant? direct effects of low-to-moderate co2 concentrations on human decision-making performance. *Environmental health perspectives*, 120(12):1671–1677, 2012.
- [7] World Health Organization (WHO). Coronavirus disease (covid-19): How is it transmitted?, 2021.